

Supplement 11-Exact

Determine whether the following equations are exact. If they are exact, solve them using the method described in class.

1. $(y + e^x)dx + xdy = 0$
2. $(3x^2 + 3y)dx + (2y + 3x)dy = 0$
3. $(6x - ye^{xy})dx + (e^y - xe^{xy})dy = 0$
4. $(2xy - e^y)dx + (xe^y - x^2)dy = 0$

Solve the exact differential equation and then find the solution of the Initial Value Problem.

1. $y^3 dx + (3xy^2 - 1)dy = 0; y(-2) = 5$